Simulated Pension fund optimization model

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Background

Pension funds are long-term investment institutions with the primary purpose of supporting people after retirement. Achieving this is not as simple as investing in assets with the highest returns. They must consider different regulatory restrictions, avoid putting too many resources into a single asset and figure out a way of diversifying the portfolio, so that both the long-term return and the level of risk remain acceptable. Because of this, we can look at pension fund management as a mathematical optimization problem, since it requires balancing risk and return while trying to satisfy many constraints.

Our project was **inspired** by the Georgian pension fund's regulatory constraints and monthly reports, in which they report their investment strategies, activities and the contents of their portfolios. They emphasize the importance of diversification, investment in domestic assets and keeping a stable long term real return. Based on these ideas, we created a simplified pension fund optimization model with historical data taken from a diverse set of equities and bonds. In it, we included several realistic constraints such as minimum investment in domestic assets, maximum investment in individual ones, limits on share of bond and equity in portfolio, and minimum required return to account for inflation. While the real world pension fund management includes many more factors and complex decisions, our model shows a way of how linear mathematical optimization can be applied to portfolio allocation problems under realistic financial and investment requirements.

In the given topic we saw a great opportunity to use both MBOR techniques and our understanding of economics. We tried to dive as deep as possible into regulatory constraints given our data restrictions, in order to build detailed and logical model.

Literature

Portfolio optimization is a vital topic in financial, economic and operational research. It comes from the famous Modern Portfolio Theory written by Markowitz (1952). In his work, Markowitz introduced a mean-variance framework, where investors would calculate a given level or risk with variance and then chose portfolios which would maximize expected returns. This approach highlighted the importance of diversification and gave us idea about the efficient frontier, which includes the set of all portfolios that cannot increase return without also raising risk, which we also included for our model even though we applied linear portfolio techniques such as MAD.

Despite the fact that the mean-variance model had great theoretical importance, it still has many limitations like the risk being dependent on variance and the computational challenges when trying to solve large scale quadratic optimization problems, similar to difficulties which we come across in many other programming directions which we refer to as "Curse of dimensionality". To eliminate some of these disadvantages a new model called the Mean Absolute Deviation (MAD) was proposed by Konno and Yamazaki (1991). It replaced variance as a measure of risk with average absolute deviation of portfolio returns from their mean. Obviously, this approach had one huge advantage, it transformed the portfolio optimization model into a linear programming one, which made it possible to work on large datasets while being computationally efficient and less time consuming. Of course, as technologies improve the difference in time consumed for the 2 type of approaches is becoming less significant.

Some studies show that MAD based portfolios produce similar results as mean-variance optimizations while requiring fewer assumptions about asset return distribution. Of course, following the Modern portfolio theory and MAD frameworks, significant amounts of research have been put into applying portfolio optimization models to more realistic environments and using them for large institutional investors, such as pension funds. While classical portfolio theory assumes unrestricted allocation across assets, in contrast, later studies showed that real world investment decisions are subject to various constraints that affect optimal portfolio selection. As a result, diversification constraints were considered vital.

In case of pension fund management, even stricter boundaries are needed since regulatory requirements and long-term stability must be both satisfied at the same time. These restrictions include diversification rules, limits on different sectors, bounds on asset weights and minimum return requirements. Such restrictions often lead to stable portfolios and reduce extreme concentration in individual assets, especially in large scale investment problems.

In addition to academic optimization theories there are sources focusing on practical pension fund management on institutional level. For example, Georgian pension fund provides investment policy documents where they define for different three risk types of portfolios what are allowable ranges of equities, bonds, domestic currency denominated assets and so on (Georgian Pension Fund, 2023). They also publish monthly reports where they further highlight importance of stability and positive real return (Georgian Pension Fund, 2026).

Data

The dataset we used in this study consists of daily market data obtained from **Yahoo Finance**, covering the period from May 2018 to December 2025 (approximately 7.5 years). The dataset contains daily closing prices, as well as daily high and low prices, for a diversified set of 14 financial assets. These assets are divided into four main categories: equity exchange-traded funds (ETFs), bond ETFs, Georgian-related assets, and an individual consumer sector stocks such as Coca-Cola and Intel Corporation.

We collected data about the exact equity ETF's (**IWY** is an exception), which had major share in investment of Georgian pension fund according to their monthly reports. The equity ETF group includes, **SPY** (SPDR S&P 500 ETF), **IVV** (iShares Core S&P 500 ETF), **VGK** (Vanguard FTSE Europe ETF), **VPL** (Vanguard FTSE Pacific ETF), **VWO** (Vanguard FTSE Emerging Markets ETF), **EWY** (iShares MSCI South Korea ETF), and **ILF** (iShares Latin America 40 ETF), representing the USA, developed and emerging markets equity exposure. The bonds related assets consist of **IEF** (iShares 7–10 Year Treasury Bond ETF) and **TLT** (iShares 20+ Year Treasury Bond ETF), representing medium-term and long-term U.S. government bond exposure. while **KO** (Coca-Cola Company) and **INTC** (Intel Corporation) are included as a large multinational consumer-sector equities.

The Georgian-related assets include **BOG** (Bank of Georgia Group), **TBC** (TBC Bank Group), and CGEO (Georgia Capital), which we assumed to be **"domestic"** this is one of our models strict assumptions because Georgian stock market is too small in size and we could not obtain consistent data with other relevant international assets. These three Georgian-related assets required additional preprocessing due to currency denomination differences. Although BOG, TBC, and CGEO are economically connected to Georgia, their shares are traded on the London Stock Exchange and are quoted in thousand pence. Therefore, their prices were first converted into British pounds and then converted into U.S. dollars using historical daily GBP/USD exchange rates.

Model

We formulated our simulated pension fund portfolio allocation problem as a constrained optimization model where objective is to minimize Mean Absolute Deviation (MAD), subject to some relevant real life institutional and practical investment constraints. It is worth to mention that we are considered and were inspired by the regulatory caps, which we adjusted according to our dataset limitations, for example we only have three assets considered as domestic Georgian ones and found consistent data about other 11 assets (5 of the ETF's same as Georgian pension fund most massively invested ones). If we take into account that we have only equities and only 2 bond ETF's (Fixed income securities) and 3 domestic sources our model could be most comparable to highest risk Georgian pension fund portfolio where **Fixed income securities** are limited at **50%** equities range from **40%** to **60%** and foreign equities are capped at 60% (Georgian Pension Fund, 2023). As for bonds, we had to effectively tighten limit to 24% since we had only 2 ETF's and we had diversification concerns (Actually we capped each bond ETF at 12%). And due to data limitation, we ended up with foreign equities **implicitly** at 66% *(100%-24%(Bonds)-10%(Domestic))*. We also have some relevant diversification caps included, note that it will be slightly higher for **Bond ETF's** as they are relatively safer than **Equities**. Also, we came up with such idea: to make sure all major Georgian related assets get support we made our model to give at least **1% symbolic** investment percentage to each of them so that only one of Georgian assets does not grab all the investment dedicated for "Domestic" assets. **Decision Variables:** Domestic (Equity): BOG, TBC, CGEO; Foreign Equity: SPY, IVV, VGK, VPL, VWO, EWY, ILF, KO, INTC; Bond ETFs(non-equity): IEF, TLT.

Decision Variables and some important parameter definitions:

Let w_i denote the portfolio weight allocated to asset i we listed them above in red. The decision variables represent the proportion of total wealth invested in each asset. All weights are continuous and non-negative.

$R_{i,t}$ denotes the return of asset i in period t, and T is the total number of monthly observations. The portfolio return at time t is defined as:
$R_t^p = \sum_{i=1}^{N} w_i R_{i,t}$. The average portfolio return is given by $\bar{R}^p = \frac{1}{T}\sum_{t=1}^{T} R_t^p$.

Our model formulation

<u>**Objective**</u> - Minimize MAD: $\sum_{t=1}^{T} | \boldsymbol{R_t^p - \bar{R}^p} |$ where $R_t^p = \sum_{i=1}^{N} w_i R_{i,t}$ and $\bar{R}^p = \frac{1}{T}\sum_{t=1}^{T} R_t^p$

Constraints:

$\sum_{i=1}^{n} \boldsymbol{w_i = 1}$ *(Budget constraint)*

$\boldsymbol{w_i \leq 0.1 i} \in \{$***Equity assets***$\}$ *(Diversification constraint)*

$\boldsymbol{w_i \leq 0.12 i} \in \{$***Bond assets***$\}$ *(Diversification constraint)*

$\sum_{i \in Domestic} \boldsymbol{w_i \geq 0.1}$ *(Domestic assets minimum allocation)*

$\boldsymbol{w_{i\_D} \geq 0.01}$ *(Single domestic asset's minimum allocation, we ensured all major ones get support)*

$\sum_{i \in Bonds} \boldsymbol{w_i \geq 0.2}$ *(Minimum bond allocation)*

$\boldsymbol{(1 + \bar{R}_{pmonthly})^{12} - 1 \geq 0.04}$ *(Minimum nominal annual required return (compounded monthly))*

$\boldsymbol{w_i \geq 0}$ *(no short selling, that's because Penson funds usually prioritize long term gains)*

Results

Analyzing optimal weights

Asset	Weight
TLT	0.120000
IEF	0.120000
VGK	0.100000
IVV	0.100000
KO	0.100000
SPY	0.100000
VWO	0.100000
VPL	0.100000
CGEO	0.051592
ILF	0.049998
TBC	0.038408
INTC	0.010002
BOG	0.010000
EWY	0.000000

The results of the baseline model is displayed on the left, as you can see our pension fund optimal proportions are primarily driven by diversification constraints. All the exchange-traded funds hit the threshold and the bond ETF's also hit the threshold though their cap was slightly larger since bonds tend to be less risky than equity assets. Interestingly though Coca Cola corporation is exactly at the cap another big company, INTC, is barely above 1%. For us, the most interesting result is division of weights between Georgia Capital, TBC and BOG. CGEO gets more than 5%, TBC around 4% and 1 % goes to Bank of Georgia. Note that all combined get exactly 10% of the portfolio, so relative to other assets "domestic" are relatively more volatile which was completely expected. Interestingly BOG got exactly 1% which correspond to our rule that pension fund had to invest a symbolical 1% in each major Georgia related companies. Expected annual **nominal** return was **7.19%.** You can also check out the pie chart **Figure A1** in the appendix.

Analyzing risk measures for the assets

Asset	Std_Dev	MAD	Std_Rank	MAD_Rank
BOG	0.12751	0.09517	1	1
TBC	0.11656	0.08336	2	2
INTC	0.11345	0.08306	3	3
CGEO	0.10659	0.07727	4	4
ILF	0.08116	0.061	5	5
EWY	0.07537	0.05831	6	6
VGK	0.05295	0.04196	7	7
KO	0.05034	0.03846	8	8
IVV	0.04857	0.03756	9	9
SPY	0.04851	0.03749	10	10
VWO	0.04754	0.03608	11	11
VPL	0.0448	0.03345	12	12
TLT	0.04208	0.0334	13	13
IEF	0.02027	0.01589	14	14

Here is a table of **MAD** (risk) also with std deviations. As you can see in relative terms both MAD and std deviation are similar. So, if we used **Markowitz** portfolio theory with **variance** as measure of **risk** we would get probably close results. As we can see **BOG** is the one with highest volatility so that's most probably why optimization setup minimizing MAD was reluctant giving BOG a meaningful share. The fact that **CGEO** got more share than **TBC** is also apparently linked with riskiness too as **TBC** is relatively riskier. Interestingly **IWY** (High level Korean stocks) got weight of 0. That could be linked with marginal returns, which could be relatively less than ones offered by other ETF's, different from Korean market.

Analyzing the shadow prices

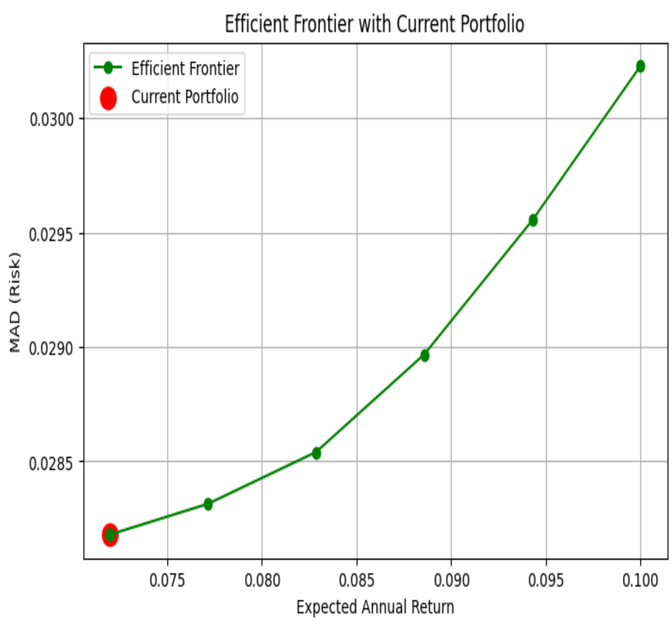
As our next step, we obtained dual variable values (the shadow prices) for each of the constraints. In total 21 values for 21 constraints apart from non-negativity ones. Interestingly what duals suggest is that, since bond ETF's which are **IEF** (7-10 years bonds) and **TLT** (20+ year treasury bonds), have shadow prices of approximately -3.60 and -3.24 respectively, which means that constraints are binding. Unfortunately, we cannot directly interpret it as marginal effect of a specific cap increase on objective function as we'd be leaving feasibility set. However, we can certainly conclude that by easing cap on **IEF (So increasing max. weight)** would bring us significant reduction in MAD (our risk measure) and easing cap on **TLT** would give second best result int that regard. So, it appears that it is logical that we gave higher cap to bond ETFs compared to equities, which at least in theory tend to be riskier on average.

As for most Equity ETF's (Excluding **EWY=0 and ILF=0 since their caps are non-binding**) and **KO** (Coca Cola) as they also have negative values ranging from -1.93 to -0.72, we see that those constraints once again are binding and our model prefers the cap to be increased to further reduce risk. In this setup we would benefit most from easing (By easing we mean increasing because we have ≤ constraint) cap on Coca cola whose shadow price value is -1.93 and belongs to equities.

It is interesting to note that Domestic constraint cap has positive shadow price of ~0.376, so it is binding (our model picked the least allowable domestic investment overall) and by increasing it, requiring higher domestic investment, we would increase overall riskiness of the portfolio. Furthermore, minimum investment of 1% cap for BOG had positive value which means that by hypothetically increasing it with even small value we are increasing risk, hence our model prefers to keep it as low as possible. Other caps not mentioned here apart from the budged constraint all had 0 values since as it was visible from the optimal weights too, they were not binding.

Sensitivity analysis results & robustness check, worst & best-case scenarios & efficient frontier

Furthermore, we did sensitivity analysis for 2 following caps we came up with: Minimum required investment in domestic assets and minimum required return. We tested our main model with different minimum requirements for investing in the given direction for proportions ranging from 0.5 to 0.25 and saw that decreasing the requirement to 0.05 only reduced MAD by 0.0001 units and increasing to 0.15 increased risk measure by 0.0003, further +5% jumps made increase even more rapid at 0.25 increase in MAD was already 0.0024, so we were convinced that leaving as it was a sensible decision. As for the minimum required return, as we read in Georgian pension fund monthly reports they aim at positive real return, so in our case considering expected highest inflation of around 3.8% for USD annually we set minimum required return to 4% (nominal) annually. Actually, it turned out that it wasn't binding and model was giving us nominal 7.19% expected return per year. You can check out **Figure A2** in the Appendix showing MAD versus Minimum required rate indicating where the requirement becomes binding (*Risk increases*).



We also did robustness checks and took the dataset with lowest daily prices and ran the same model and repeated the same process for highest daily prices too. Interestingly, the ones who hit the caps were still represented the same way, but others had some changes for instance Intel corporation whose weight increased to **2.4% in "worst"** scenario and decreased to **0.46% in "best"** scenario. Indicating that in relative terms some assets perform better in "worst" conditions. Check out **Table 1A** in appendix for more details.

Here is the efficient frontier for our model on the left. As we can see we are at efficient point which is at the same time minimum risk one with a return of **7.19%** approximately. This is not surprising as our minimum required return constraint is non-binding, if it was binding then we would be moving higher across the frontier with higher return but increased risk. This actually highlights the importance of regulatory floors in shaping portfolio risk-taking behavior, as they act as primary mechanism that would move us away from the minimum risk corner to a point located higher up on the efficient frontier.

Given that expected inflation is at most 3.8% for US, we'd **expect ~ 3.26% of real return** next year, which is calculated as follows: (1+0.0719) / (1+0.038) - 1.

Conclusion

Asset	Weight	Allocation (USD)
INTC	0.010002	$10,002
KO	0.1	$100,000
SPY	0.1	$100,000
IVV	0.1	$100,000
VGK	0.1	$100,000
VPL	0.1	$100,000
VWO	0.1	$100,000
EWY	0	$0
ILF	0.049998	$49,998
IEF	0.12	$120,000
TLT	0.12	$120,000
BOG	0.01	$10,000
TBC	0.038408	$38,404
CGEO	0.051592	$51,592

We can conclude that if we were to use our model in a hypothetical scenario where a pension fund has to decide how an additional 1 million USD should be invested, the optimal investment expenditure per assets we used would be following: 10,000$ in BOG, 38,404$ in TBC and 51,592$ in Georgia capital. So, in total 100,000$ in "domestic" assets. Another 900,000$ would be invested in foreign assets, of which 240,000$ would be bonds ETF's and the rest equity ETFs combined with Coca-Cola and Intel Corporation. As you can see the details are displayed on the table on the left.

As for conclusion of the model results, they indicate that optimal portfolio, to considerable extent, is shaped by regulatory constraints that we incorporated. This fact was expected as we had to work with real world pension fund regulations which tend to be strict in order to maintain high diversification and minimize loss from unexpected shocks world-wide. Furthermore, from a risk point of view for most of the cases our model showed that assets with higher volatility get lower weight assigned by our model in the portfolio. By investigating duals, we proved that significant number of constraints were binding, which reveals that regulations in some directions play bigger role diversification rather than pure risk minimization with MAD, in many cases our model preferred less strict diversification caps. Robustness checks also showed that there was no significant change in worst case scenario which again supports the importance of regulatory constraints alongside with purely mathematical minimization of the risk measure. More importantly, we managed to get positive real return.

Potential future work possibilities we noticed:

Furthermore, when we compared standard deviation as risk measure and MAD, they ranked the assets from most risky to least risky ones in the same order. We believe for **future work** it would be interesting to explore differences between MAD and classical Markowitz VAR minimization outcomes. Especially, how VAR as risk measure would change the weights, and whether it assigns similar weights or at least asset weights relative to each other remain similar as well.

Another relevant extension could potentially be adding results and interpretation for CVaR model. Unlike MAD and variance-based approaches this one would focus on minimizing expected losses in the worst-case scenario, which would could be also relevant for pension funds, given that we come up with specific scenarios.

Finally, if we aimed at more closely simulating Georgian pension fund portfolio optimization, in terms of data used it would be better for future works to obtain Georgian stock market data and bonds which are denominated in GEL and introduce constraints related to currency denomination as well. Of course, given that we increase domestic assets we can think of including wider range of foreign equities and bonds. This would enable us to use closer numbers for domestic or foreign asset allowed proportions like Georgian pension fund policy stated, or equity and bonds proportions. Coming up with different models with where constraints differ for low, medium and high-risk categories of pension fund portfolios and their results comparison would also be meaningful.

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NOTE: *Appendix on the next 8th page*

Appendix A: Additional tables and figures

Figure A1: Pie chart showing the distribution of portfolio weights:

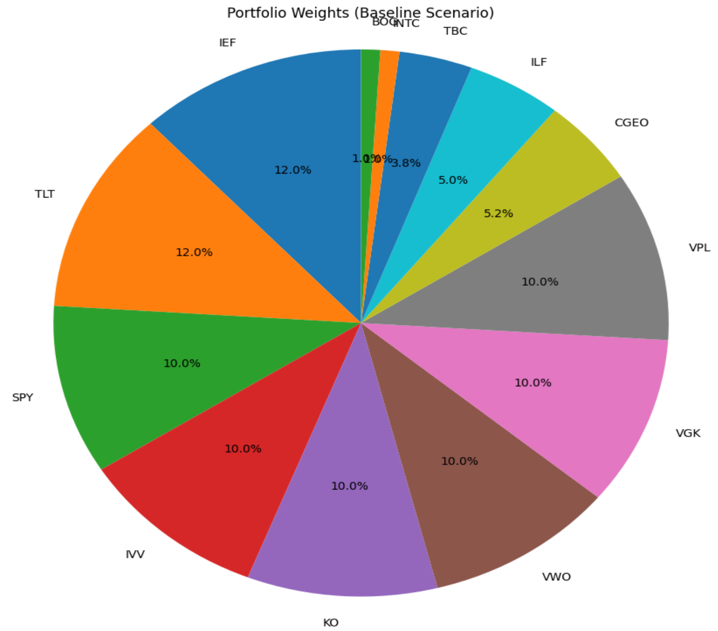
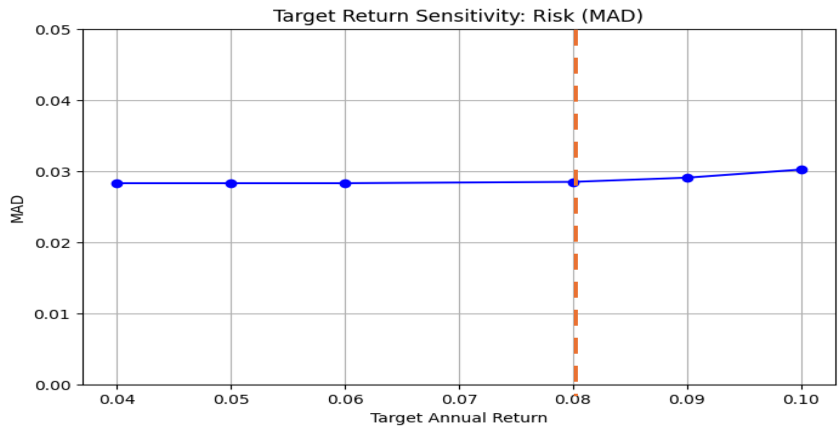


Figure A2: Target return sensitivity



MAD rises roughly from 8% annual rate after it becomes binding. (Because it was tested at 8% point though it more precisely has to increase from ~ 7.2%)

Table A1 Comparison of Baseline, Worst- & Best-case optimal weights

Asset	Baseline	Low Scenario	High Scenario
TLT	0.120000	0.120000	0.120000
IEF	0.120000	0.120000	0.120000
VGK	0.100000	0.100000	0.100000
IVV	0.100000	0.100000	0.100000
KO	0.100000	0.100000	0.100000
SPY	0.100000	0.100000	0.100000
VWO	0.100000	0.100000	0.100000
VPL	0.100000	0.100000	0.100000
CGEO	0.051592	0.050400	0.051807
ILF	0.049998	0.035855	0.055414
TBC	0.038408	0.039600	0.038193
INTC	0.010002	0.024145	0.004586
BOG	0.010000	0.010000	0.010000
EWY	0.000000	0.000000	0.000000

Below VPL you can notice apparent changes of assets allocation